EE3621: Digital Signal Processing | III B.Tech EEE

Faculty Reference Document — Textbook Replacement

PREFACE FOR FACULTY

This lecture covers the **Matched Z-Transform (MZT)** and **Digital-to-Digital Spectral Transformations (Constantinides Transformations)**. Spectral transformations allow an engineer to design a simple discrete lowpass prototype and convert it directly into highpass, bandpass, or bandstop filters by algebraic variable substitution.

Pedagogical Strategy:

1. Formulate the Matched Z-Transform (MZT): Direct pole-zero mapping $(s - s_k) \to (1 - e^{s_k T_d} z^{-1})$.
2. Address the MZT zero-placement rule: Adding factors of $(1 + z^{-1})$ to enforce zero gain at half the sampling frequency ($\omega = \pi$).
3. Formulate the **Constantinides Digital Spectral Transformations**:
   - Lowpass to Lowpass: $z^{-1} \to \frac{z^{-1}-\alpha}{1-\alpha z^{-1}}$
   - Lowpass to Highpass: $z^{-1} \to -\frac{z^{-1}+\alpha}{1+\alpha z^{-1}}$
   - Lowpass to Bandpass & Bandstop substitutions.
4. Prove that all-pass spectral mappings map the unit circle to the unit circle and preserve stability.

1. LEARNING OBJECTIVES

By the end of this lecture, students will be able to:

1. **Apply** the Matched Z-Transform to convert analog transfer functions to discrete transfer functions.
2. **Transform** lowpass digital filter prototypes into highpass, bandpass, and bandstop filters using Constantinides formulas.
3. **Calculate** the mapping parameter α from prototype and target cutoff frequencies.

2. MATHEMATICAL FOUNDATIONS

2.1 The Matched Z-Transform (MZT)

Given analog factored transfer function:

$$H_a(s) = K \frac{\prod_{i=1}^{M} (s - z_i)}{\prod_{k=1}^{N} (s - p_k)}$$

The discrete transfer function is formed by mapping each factor:

$$H(z) = K_d \frac{\prod_{i=1}^{M}(1 - e^{z_i T_d} z^{-1})}{\prod_{k=1}^{N}(1 - e^{p_k T_d} z^{-1})}(1 + z^{-1})^{N-M}$$

Where K_d is chosen to match the DC gain or passband peak.

2.2 Constantinides Digital Spectral Transformations

Substitute $z^{-1} \to g(z^{-1})$ into prototype lowpass filter $H_{\text{LP}}(z)$ with cutoff θ_p:

1. **Lowpass to Highpass (Target Cutoff ω_p):**

$$z^{-1} \to -\frac{z^{-1} + \alpha}{1 + \alpha z^{-1}}, \qquad \alpha = -\frac{\cos\left(\frac{\theta_p + \omega_p}{2}\right)}{\cos\left(\frac{\theta_p - \omega_p}{2}\right)}$$

2. **Lowpass to Lowpass (New Cutoff ω_p):**

$$z^{-1} \to \frac{z^{-1} - \alpha}{1 - \alpha z^{-1}}, \qquad \alpha = \frac{\sin\left(\frac{\theta_p - \omega_p}{2}\right)}{\sin\left(\frac{\theta_p + \omega_p}{2}\right)}$$

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3. WORKED NUMERICAL EXAMPLES

Example 29.1: Lowpass to Highpass Spectral Transformation

Problem: A first-order digital lowpass filter prototype has transfer function:

$$H_{\text{LP}}(z) = \frac{0.2452(1 + z^{-1})}{1 - 0.5095 z^{-1}}$$

with cutoff $\theta_p = 0.2\pi$. Transform it into a digital Highpass filter with cutoff $\omega_p = 0.6\pi$.

Solution:

1. **Compute Transformation Parameter α:**

$$\alpha = -\frac{\cos\left(\frac{0.2\pi + 0.6\pi}{2}\right)}{\cos\left(\frac{0.2\pi - 0.6\pi}{2}\right)} = -\frac{\cos(0.4\pi)}{\cos(-0.2\pi)} = -\frac{\cos(72^\circ)}{\cos(36^\circ)} = -\frac{0.3090}{0.8090} = -0.3820$$

2. **Substitute $z^{-1} \to -\frac{z^{-1} - 0.3820}{1 - 0.3820 z^{-1}}$:**

$$H_{\text{HP}}(z) = \frac{0.2452\left(1 - \frac{z^{-1} - 0.3820}{1 - 0.3820 z^{-1}}\right)}{1 - 0.5095\left(-\frac{z^{-1} - 0.3820}{1 - 0.3820 z^{-1}}\right)} = \frac{0.2452[(1 - 0.3820 z^{-1}) - (z^{-1} - 0.3820)]}{(1 - 0.3820 z^{-1}) + 0.5095(z^{-1} - 0.3820)}$$

$$H_{\text{HP}}(z) = \frac{0.2452[1.3820 - 1.3820 z^{-1}]}{(1 - 0.1946) + (-0.3820 + 0.5095) z^{-1}} = \frac{0.3389(1 - z^{-1})}{0.8054 + 0.1275 z^{-1}} = \frac{0.4208(1 - z^{-1})}{1 + 0.1583 z^{-1}}$$

3. **Verification:**
   - At DC ($z = 1$): $H(1) = 0$ (Zero transmission at DC).
   - At $\omega = \pi$ ($z = -1$): $H(-1) = \frac{0.4208(2)}{1 - 0.1583} = \frac{0.8416}{0.8417} \approx 1.0$ (0 dB passband gain).

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4. UNIVERSITY EXAMINATION QUESTIONS & MARKING RUBRIC

Question 1 (15 Marks)

(a) State the formulas for digital-to-digital spectral transformations from Lowpass to Highpass and Lowpass to Bandpass. Explain why these transformations preserve system stability. *(8 Marks)* **(b)** Given a 1st-order lowpass filter $H(z) = \frac{1-a}{2}\frac{1+z^{-1}}{1-az^{-1}}$ with 3-dB cutoff $\theta_p = \pi/2$, find a and transform it into a Highpass filter with cutoff $\omega_p = 3\pi/4$. *(7 Marks)*

Model Answer & Step-by-Step Marking Rubric:

- **Part (a):**
  - Transformation equations for LP $\to$ HP and LP $\to$ BP *(4 Marks)*
  - Stability proof: The mapping $g(z^{-1})$ is an all-pass function ($|g(e^{j\omega})| = 1$), mapping $|z| < 1$ strictly to $|z| < 1$ *(4 Marks)*
- **Part (b):**
  - Find a: At $\theta_p = \pi/2$, $|H(j)|^2 = 0.5 \implies a = 0 \implies H_{\text{LP}}(z) = \frac{1+z^{-1}}{2}$ *(3 Marks)*
  - $\alpha = -\frac{\cos(5\pi/8)}{\cos(-\pi/8)} = -\frac{-0.3827}{0.9239} = 0.4142$ *(2 Marks)*
  - Substitute to find $H_{\text{HP}}(z) = \frac{1-0.4142}{2}\frac{1-z^{-1}}{1+0.4142z^{-1}} = 0.2929\frac{1-z^{-1}}{1+0.4142z^{-1}}$ *(2 Marks)*

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5. PYTHON VERIFICATION SCRIPT

```
import numpy as np

# Verify HP transformation
w = np.array([0, np.pi])
H_hp = lambda z: 0.4208 * (1 - z**-1) / (1 + 0.1583 * z**-1)
print("Gain at w=0 (DC):", np.abs(H_hp(np.exp(1j * 0))))
print("Gain at w=pi:", np.abs(H_hp(np.exp(1j * np.pi))))
```